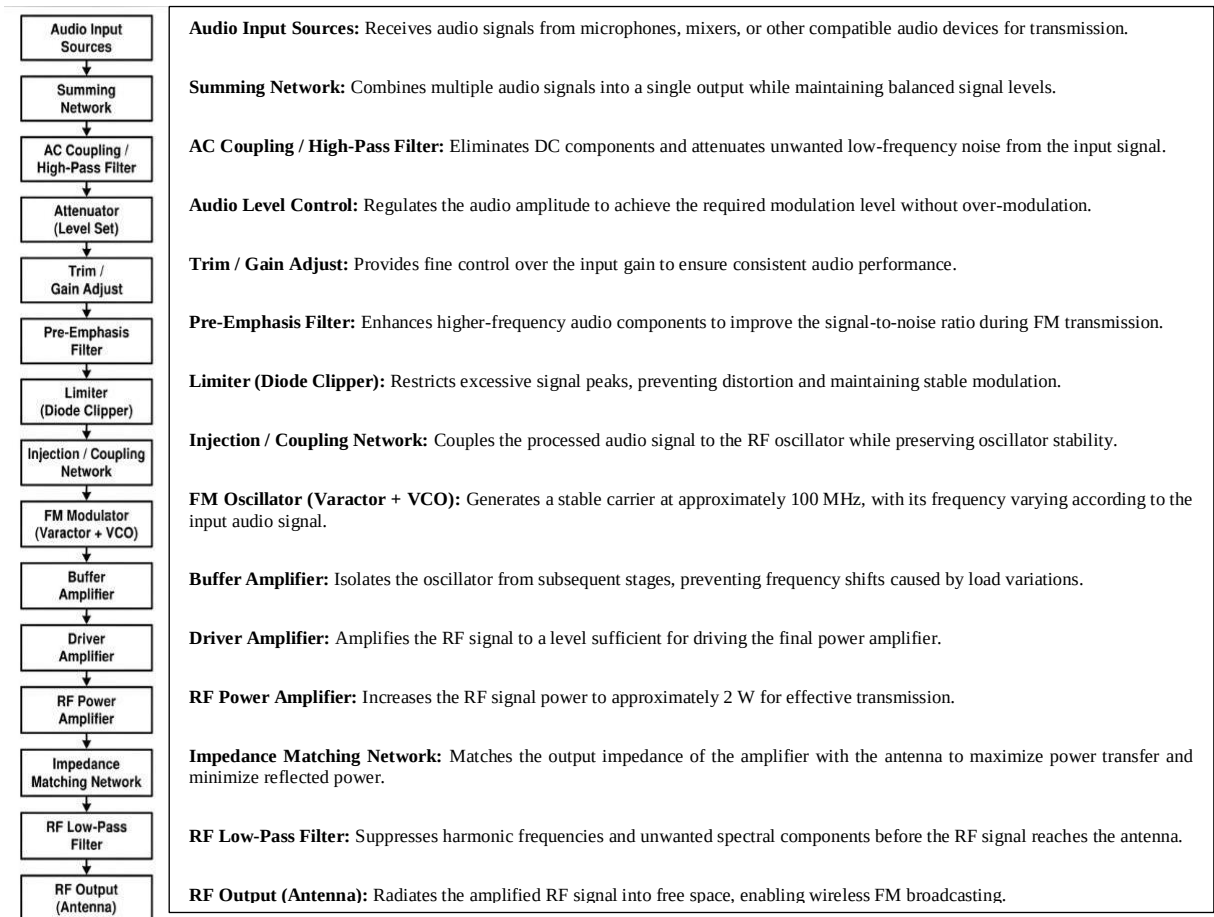
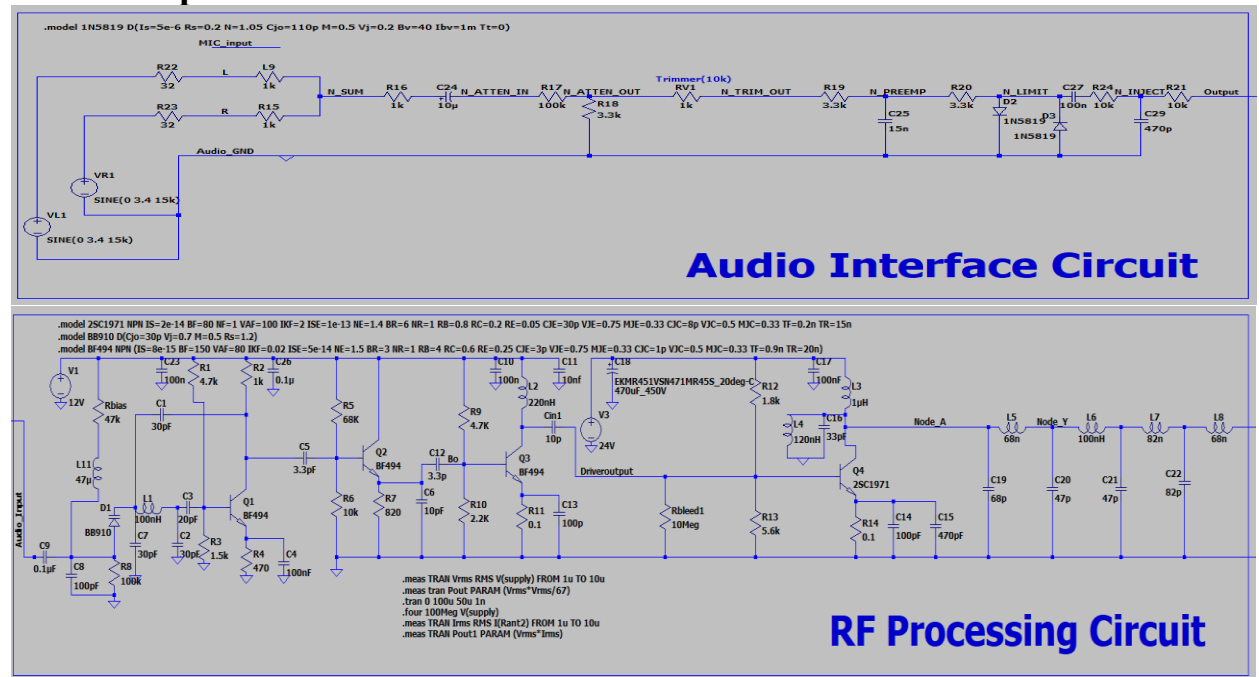


FM Radio Transmitter

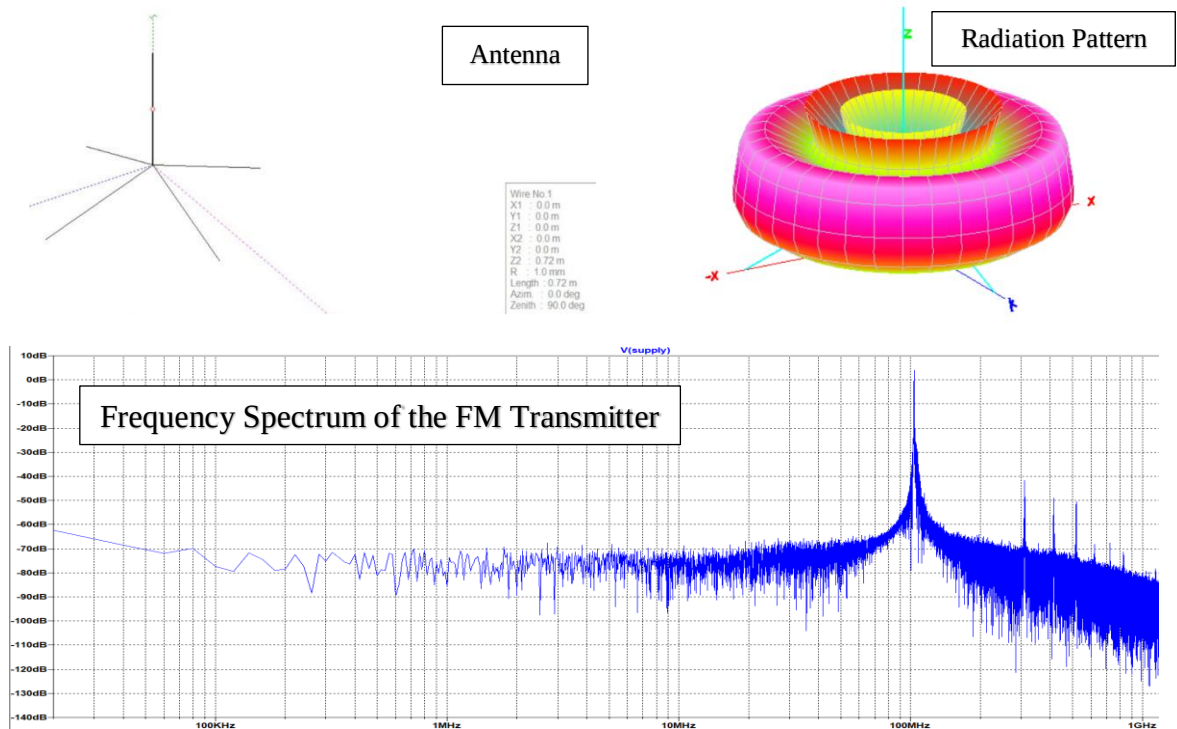
System Workflow



Circuit Implementation



Simulation Results



The complete transmitter and antenna system were verified through circuit and electromagnetic simulations. LTspice analysis confirmed stable operation of the RF transmitter with a carrier frequency of approximately **100 MHz**, an estimated RF output power of **2.4 W**, and a **Total Harmonic Distortion (THD)** of approximately **3.18%**, indicating acceptable spectral performance. Antenna simulations demonstrated effective impedance matching with an **SWR of approximately 1.34**, a feed-point impedance of **66.85 – j0.18 Ω**. The obtained radiation pattern and matching characteristics indicate that the antenna is well suited for efficient operation in the FM broadcast band.